

Personal Statement and Project Proposal

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PERSONAL STATEMENT

My path into computational biology started with baseball. During my Summit award project this past spring, I have been building an XGBoost model to predict which NCAA baseball players would get drafted into the MLB. That project taught me how to wrangle messy, real-world datasets, and interpret model outputs critically. But it also showed me something bigger: that the same machine learning toolkit I was using to predict draft picks could be applied to problems with far greater stakes. That realization is what pulled me toward disease prediction.

When I first encountered neural networks in my coursework (CS4100 and DS4420), I was struck by how naturally they represent relationships; not just between data points, but between people, places, and the ways disease actually moves through a city. Boston is a city of neighborhoods, each with its own demographics, transit patterns, and health infrastructure. The idea that you could model those neighborhoods as nodes in a network, connected by the subway lines and bus routes that residents actually use, felt like a genuinely better way to think about disease forecasting than treating each neighborhood as an independent data point.

My previous research experience has given me a solid foundation for this work. As a Data Research Developer for Prof. Grodal and Prof. Chang, I wrote Python pipelines to process over 10,000 archival documents, which taught me how to automate data cleaning and handle messy, inconsistent sources. And my work in Dr. Jennings' project, where I built automated pipelines for converting MRI scans into finite element models, reinforced the importance of writing reliable, well-documented code. Beyond machine learning fundamentals, I have also spent time this semester doing preliminary work for this GNN project through my Research Practicum course (CS4545). I have been collecting and cataloging datasets, mapping naming conventions across fourteen neighborhoods, and working through the literature on graph neural networks for disease forecasting. That preparation has given me a strong grasp of both the data landscape and the methodological approach before the summer even begins.

I also learned, sometimes the hard way, that research does not follow a straight line. When my MLB draft model underperformed compared to professional scout reports, a Spearman correlation of 0.704 versus their 0.837, I had to step back and recognize that I was missing an entire category of data. High school players, which scouts evaluate in person, had no standardized data available online for me to collect. Rather than forcing a flawed approach, I reframed the problem to focus specifically on NCAA players and shifted from predicting exact pick numbers to modeling the relative order of college

draftees. I was able to learn that in research there are multiple paths that can be taken, but make sure to have the proper data before you go down one.

I anticipate needing that same adaptability this summer. I am working with data from multiple public agencies, and I have already experienced some difficulty getting consistent responses from BPHC when requesting additional data. I have also taken on mentoring a classmate, Atharva, on this project, which means I need to balance my own learning with bringing someone else up to speed and integrating his contributions meaningfully. These are real challenges, but I see them as part of what makes independent research valuable: learning to navigate not just technical problems, but logistical and collaborative ones too.

This summer, I hope to gain deeper expertise in deep learning architectures, specifically graph neural networks and how they handle spatial and temporal data simultaneously. I also want to develop my ability to communicate with public health stakeholders and understand what makes a model not just accurate but actually useful in a decision-making context. There is a big difference between a model that performs well on a test set and one that a public health official can actually use to decide where to send a mobile vaccination clinic next week.

I aim to apply for a Computational Biology PhD after I graduate, and I believe the PEAK Trail Blazer Fellowship is not only good for research experience, but also understanding more of how I like working with traditional and nontraditional computational biology methods. I also hope this fellowship leads into real world impact on Boston's public health response of influenza, and potential work experience in the Public Health field.

PROJECT PROPOSAL

Project Background, Significance, and Objectives

Seasonal influenza imposes an average annual economic burden of \$11.2 billion in the United States, with direct medical costs of \$3.2 billion and indirect costs of \$8.0 billion from lost productivity [7]. The disease causes approximately 26.2 million cases, 247,000 hospitalizations, and 36,300 deaths each year. Despite this substantial burden, local health departments lack tools to predict which neighborhoods will experience outbreaks one to two weeks in advance. Current population-wide vaccination recommendations provide general guidance [2], but do not support the kind of localized, proactive actions that could reduce transmission in high-risk communities. Because influenza vaccines require approximately two weeks to develop protective immunity, interventions must precede outbreaks rather than react to them.

Recent work has applied graph neural networks to disease forecasting with promising results. Luo et al. (2025) demonstrated that a GNN that captures both geographic proximity and functional connections between regions could improve the precision of influenza prediction in Taiwan [5]. However, that model took a data-minimalist approach: it used only influenza-like illness rates at each node and inferred connectivity through statistical correlations in case patterns rather than actual mobility data. This philosophy was informed in part by cautionary lessons from Google Flu Trends, which initially showed that search query data could estimate ILI rates one to two weeks faster than CDC surveillance [1], but failed systematically after 2011 due to algorithm changes and overfitting: a phenomenon termed "big data hubris" [4]. The data-minimalist response to that failure was understandable, but it also limits the information available for prediction and has operated primarily at regional or county scales focused on accuracy benchmarks rather than actionable public health decisions. Meanwhile, work on urban mobility and disease transmission has shown that transit networks play a significant role in how disease moves through cities. Malik et al. (2022) used New York City subway ridership as a proxy for overall mobility to improve COVID-19 forecasting [6], demonstrating that mobility-dependent infection rates better captured outbreak dynamics than standard models. Agent-based simulations have explored transmission within transit systems themselves [3]. However, these approaches have either operated at coarse spatial scales, focused on retrospective fitting rather than operational forecasting, or modeled theoretical scenarios with uniform populations.

This project proposes a conceptual inversion of the data-minimalist approach. Where the Taiwan model inferred relationships from statistical correlation, we construct them from actual MBTA transit flows and geographic adjacency. Where it used a single feature per node, we use a multi-dimensional feature vector incorporating wastewater surveillance, emergency department visits, demographics, and socioeconomic data. The objective is to develop a GNN model that predicts influenza cases at the neighborhood level in Boston, with forecasts one to two weeks ahead of current surveillance, specifically designed to support decisions like where to position mobile vaccination clinics, where to intensify community outreach, and where to deploy targeted incentive programs for vaccines before transmission accelerates.

Project Design and Methodology

We represent Boston as a graph of 14 neighborhoods defined by the Boston Public Health Commission and City of Boston Planning Development: Allston/Brighton, Back Bay–Beacon Hill–Downtown–North End–West End, Charlestown, Dorchester, East Boston, Fenway, Hyde Park, Jamaica Plain, Mattapan, Roslindale, Roxbury, South Boston, South End, and West Roxbury. Each neighborhood is a node in the graph. We construct two types of edges. Geographic edges connect neighborhoods based on spatial adjacency, with weights reflecting distance — adjacent neighborhoods receive weight 1, neighborhoods separated by one area receive weight 2, and so on up to about 3 or 4. Functional edges use MBTA ridership flows, where the edge weight from neighborhood i to neighborhood

j is proportional to the average weekly commuters traveling between them, normalized by total outflow from neighborhood i . This captures human movement patterns rather than infer statistical correlations.

Each node carries a feature vector built from three data sources standardized to weekly resolution. From the Boston Public Health Commission: neighborhood-level flu rates per 100,000 residents, ILI emergency department visit counts and percentages, and wastewater viral concentrations. From the Massachusetts Department of Public Health: vaccination rates by age group across three flu seasons (2023–2026). From the City of Boston Planning Development Agency: population density, age distribution, poverty rates, race and ethnicity, commute mode shares, housing tenure, and vehicle availability.

City-wide covariates (overall vaccination uptake, weather data, and demographic case distributions), are incorporated through a fusion layer that conditions neighborhood-level predictions on broader trends. Inspired by performance gains observed from adding a background node in the Taiwan model, we include a spatial anchor node representing areas outside Boston's core (the harbor, surrounding suburbs), connected to boundary neighborhoods like East Boston, Charlestown, and West Roxbury. This node is initialized with neutral feature values, allowing the model to recognize geographic edges of the city without introducing artificial disease signals.

The model architecture follows a pipeline: We train using mean squared error loss with L2 regularization, with data from 2018–2022 for training, 2022–2023 for validation, and 2023–2024 for testing to prevent data leakage and mimic operational forecasting conditions.

We will establish baselines using sklearn models that treat each neighborhood independently, then evaluate the GNN against these using RMSE, MAE, and per-neighborhood correlation. Ablation studies will quantify the contribution of each data stream by systematically removing components like wastewater, mobility edges, or the background node. We will also test model robustness under simulated conditions of delayed and incomplete reporting, which reflects the reality of public health surveillance.

The primary resources required are computational infrastructure for training GNNs (which I have access to) and continued data access from public agencies. One anticipated challenge is obtaining additional data from the Boston Public Health Commission, as I have already experienced some inconsistency in their responses. The model architecture is flexible enough to proceed with reduced feature sets if certain data streams prove unavailable, and results will document the impact. I am also mentoring a classmate, Atharva, on the project and will onboard him progressively through the data collection, preparation, and graph construction phases.

Mentoring Plan

I will meet with my faculty advisor weekly via online meetings on Teams, with a target of one hour per session. These meetings will focus on reviewing progress, discussing technical decisions (such as model architecture choices, fusion layer design, and evaluation strategy), and planning the next

week's work. Between meetings, I will communicate with my advisor via email for quick questions and share ongoing progress through a shared project repository where code, data documentation, and draft paper sections will be regularly updated. I will also coordinate with Atharva's mentor to ensure alignment on his contributions, likely through a brief joint check-in every week.

Outcomes Statement

The primary product of this project will be a research paper suitable for submission to a Public Health peer-reviewed venue or a machine learning for health workshop at a conference. The paper will present the full methodology, results, and public health implications of the GNN forecasting model, building on the related works, introduction, and proposed solution sections I have already drafted this semester. I will also produce a public GitHub repository containing the complete codebase, data processing pipeline, and trained model with documentation sufficient for reproducibility. I will create a presentation and poster for sharing the work at Northeastern through RISE. If the model demonstrates strong predictive performance, I intend to share findings directly with the Boston Public Health Commission as a potential decision-support tool for future flu seasons.

References

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